E-Commerce Sales Analysis Using NumPy

Practice Project

Skills Demonstrated

- NumPy Arrays
- Statistical Analysis
- Revenue Analysis
- Boolean Filtering
- Product Ranking
- Revenue Forecasting

Aditya Hatzade

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Problem Statement

An e-commerce company wants to analyze product sales performance and revenue generation across different product categories. The company needs insights into top-performing products, low-performing products, revenue distribution, and future growth opportunities.

Objective

The objective of this project is to use NumPy to analyze product sales data, calculate revenue metrics, identify top and low-performing products, measure revenue variability, and forecast future revenue growth.

```
# Import required library
import numpy as np

#creating numpy arrays for products, units sold, and price
products = np.array([
    "Laptop",
    "Mobile",
    "Headphones",
    "Keyboard",
    "Monitor",
    "Mouse",
    "Speaker",
    "Tablet"
])

units_sold = np.array([
```

```
    120,
    250,
    180,
    90,
    110,
    300,
    80,
    140
])

price = np.array([
    50000,
    20000,
    3000,
    1500,
    12000,
    800,
    4000,
    25000
])
```

The dataset contains product names, units sold, and selling prices for various products sold through an e-commerce platform.

Revenue Statistics

```
revenue = units_sold * price
print("Total Revenue:", np.sum(revenue))
print("Average Revenue per Product:", np.mean(revenue))
print("Median Revenue:", np.median(revenue))
print("Maximum Revenue Product:", products[np.argmax(revenue)])
#argmax returns the index of the maximum value in the revenue array,
which is then used to get the corresponding product name from the
products array.

print("Minimum Revenue Product:", products[np.argmin(revenue)])

Total Revenue: 17055000
Average Revenue per Product: 2131875.0
Median Revenue: 930000.0
Maximum Revenue Product: Laptop
Minimum Revenue Product: Keyboard
```

This section provides key revenue metrics including total revenue, average revenue, median revenue, maximum and minimum to understand overall sales performance.

Revenue Distribution Analysis

Variance and standard deviation are used to measure the spread and variability of product revenues.

```
print("Variance of Revenue:", np.var(revenue))
print("Standard Deviation of Revenue:", np.std(revenue))
```

```
Variance of Revenue: 4887887109375.0
Standard Deviation of Revenue: 2210856.6460480876
```

Revenue-Based Product Filtering

```
high_revenue = revenue > 700000
print("Products with Revenue greater than 7 Lakh:\n",
products[high_revenue])
Low_revenue = revenue < 200000
print("Products with Revenue less than 2 Lakh:\n",
products[Low_revenue])
```

```
Products with Revenue greater than 7 Lakh:
 ['Laptop' 'Mobile' 'Monitor' 'Tablet']
Products with Revenue less than 2 Lakh:
 ['Keyboard']
```

Products are filtered based on revenue thresholds to identify high-performing and low-performing products.

- We find the products with revenue higher than 7 lakhs and
- The products with revenue lower than 2 lakh.

Future Revenue Projection

```
# simulate a 10% increase in revenue over the next 5 years
years = 5
future_revenue = revenue * (1.10 ** years)
print("Projected Revenue over 5 years:\n", future_revenue)
```

```
Projected Revenue over 5 years:
 [9663060.   8052550.    869675.4   217418.85 2125873.2   386522.4
  515363.2  5636785.  ]
```

- A 10% annual growth rate is assumed to estimate future revenue over the next five years.
- Helps in sales planning.

Product Ranking Analysis

```
np.sort(revenue)
# top 3 revenue generating products, argsort returns the indices that would sort the revenue array,
#  and we take the last three indices to get the top 3 products.
top3= np.argsort(revenue)[-3:] #[-3:] gives us the indices of the top 3 revenue values.
print("Top 3 Revenue Generating Products:\n", products[top3])

Top 3 Revenue Generating Products:
 ['Tablet' 'Mobile' 'Laptop']
```

Products are ranked based on revenue generated to identify the strongest contributors to overall business performance.

Project Summary

Key analyses performed in this project include:

- Calculated product-wise revenue using units sold and product prices.
- Computed important revenue statistics such as total revenue, average revenue, and median revenue.
- Identified the highest and lowest revenue generating products using NumPy indexing functions.
- Analyzed revenue distribution using variance and standard deviation.
- Filtered products based on revenue thresholds to identify high and low performing products.
- Projected future revenue assuming a 10% annual growth rate.
- Ranked products by revenue contribution and identified the top revenue-generating products.

Business Impact

- Helps identify products contributing the most revenue.
- Assists in inventory planning and stock management.
- Supports data-driven marketing decisions.
- Highlights underperforming products for improvement.
- Provides revenue forecasts for future planning.

Conclusion

Using NumPy, the sales dataset was analyzed to calculate revenue metrics, identify top-performing products, evaluate revenue variability, filter products based on performance, rank products by revenue contribution, and project future revenue growth. The analysis demonstrates how NumPy can be used to perform efficient business analytics on structured sales data.